8 Features of manufacturing industries

8.3 Modern manufacturing methods and systems when designing for manufacture for small, medium and large-scale production

Topic	What students need to learn:
8 Features of manufacturing industries	8.3 – Characteristics, processes, application, advantages and disadvantages and the importance of considering accuracy of production and efficiency of modern manufacturing methods and systems when designing for manufacture for small, medium and large scale production: *a) production scheduling and production logistics* *b) robotics in production – robots on fully-automated production and assembly lines/cells*

Topic	What students need to learn:
8 Features of manufacturing industries	*c) materials handling systems – automated storage and retrieval systems (ASRS), automatic guided vehicles (AGVs)* *d) flexible manufacturing systems (FMS), modular/cell production systems* *e) lean manufacturing using just-in-time (JIT) systems* *f) standardised parts, bought-in components* *g) quick response manufacturing (QRM)* *h) data integration – product data management (PDM), enterprise resource planning (ERP) systems* *i) concurrent manufacturing.*

e) Lean Manufacturing using Just-in-Time (JIT) Systems

Lean Manufacturing using Just-in-Time (JIT) Systems

The underpinning principle of ***'lean manufacturing and JIT'*** is to eliminate waste and improve the quality of production.

Just-in-Time (JIT) Manufacture

Just-in-Time (JIT) manufacture means that material or components are delivered only as required for manufacture in response to orders. It is central to much modern manufacturing and its main features include:

- *good relationship with suppliers, which are pivotal to smooth operations of the system*
- *a location with good transport infrastructure for deliveries and to reduce the environmental impact of transporting items*

Lean Manufacturing using Just-in-Time (JIT) Systems

Just-in-Time (JIT) Manufacture Cont…

- *No stockpiling of materials or finished products that might be damaged, stolen or become obsolete*
- *Reduced need for storage or warehousing*
- *Computerised stock control (ideally using Automated Storage and Retrieval Systems (ASRS)) and **Kanban** (stock monitoring indicator) systems*
- *Good management/employee relationships to ensure continuity of production*

Lean Manufacturing using Just-in-Time (JIT) Systems

Just-in-Time (JIT) Manufacture Cont…

Disadvantages:

- *JIT relies upon accurate analysis and forecasts using the right information at the right time*
- *If a manufacturer is inaccurate in their predictions – i.e. there is a rise in demand for the product – stock will be used up rapidly with very little opportunity for re-supply*
- *Production can also be held up or shut down completely if the raw materials supplier has problems fulfilling orders*

Lean Manufacturing using Just-in-Time (JIT) Systems

Lean Manufacturing

Lean Manufacturing concentrates on the following strategies to enhance the JIT approach:

- *The **reduction** of **stock** and **waste (Muda)** in all its forms, such as unnecessary inventory, motion, defects and processes.*
- *The use of **Six Sigma** (topic 12.2) to aim to eliminate defects.*
- *A philosophy of continuous improvements **(Kaizen)**.*
- *A focus on relationships with customers to improve overall value.*
- *Employing multi-skilled workers and using versatile, flexible, CNC machinery.*

Lean Manufacturing using Just-in-Time (JIT) Systems

Definition:

Muda means ***wastefulness***. Work that does not add value is divided into ***<u>two</u>*** categories:

1. *Work that is necessary but not recognized by the customer as adding value - and*
2. *Work that simply is not necessary.*

Lean Manufacturing using Just-in-Time (JIT) Systems

Lean Manufacturing Cont…

- The Toyota Production System (TPS) is regarded as the pioneering lean manufacturing system.
- It encompasses features of both the JIT and lean manufacturing systems.
- The empowerment and training of workers is particularly emphasised.
- Quality is maintained by automatically stopping production if there are ***problems (Jidoka).***

Lean Manufacturing using Just-in-Time (JIT) Systems

Definition:

Jidoka (English: ***Autonomation*** *— automation with human intelligence)*: The principle of designing equipment to stop automatically and to detect and call attention to problems immediately whenever they occur *(mechanical jidoka).*

Lean Manufacturing using Just-in-Time (JIT) Systems

The *five* key stages of Lean Manufacturing

Key Stage		Content
1	**Value**	Focus on value in the context of what the customer / end user is prepared to pay for
2	**Value Stream**	Identify how ***value-adding*** and ***non-value-adding activities*** affect efficiency throughout production including: ▪ *For value-adding activities – i.e. machining, processing, painting, assembling etc* ▪ *For non-value-adding activities – i.e. scrapping, sorting, storing, counting, moving etc*
3	**Flow**	Design processes that result in ***uninterrupted flow*** – from raw material to delivery of finished product
4	**Pull**	Design manufacturing for ***'pull'*** of product through the process as a response to demand, *rather than* ***'push'*** of raw materials into the process by producing components irrespective of whether they are needed or not
5	**Perfection (Kaizen)**	Adopt an approach that ***continually improves*** working processes

Lean Manufacturing using Just-in-Time (JIT) Systems

Pull Tools - Kanban

A lean organisation needs to be set up so that orders are ***'pulled'*** through the production system.

This can be achieved using a system known a ***Kanban***. ***Kanban*** uses ***'cards'*** or ***'containers'*** as simple visual signals to indicate when to pull materials, components or products through the production system.

The system relies on a *simple rule* of only producing or delivering when a card or empty container is passed to a workstation or manufacturing cell.

Pull Tools - Kanban

Production Kanban and ***Transportation Kanban*** are combined into one integrated system.

The ***production Kanban*** includes details of the operations that need to be carried out at the workstation or manufacturing cell.

The ***transportation Kanban*** only contains details regarding where materials, components or products have come from and where they are going to.

Lean Manufacturing using Just-in-Time (JIT) Systems

Pull Tools - Kanban

The main benefit of using Kanban is that it reduces the amount of work-in-progress and finished goods in stock.

Kanban restricts the supply of materials and components until they are needed, which provides and effective JiT system.

Lean Manufacturing using Just-in-Time (JIT) Systems

Perfection Tools - Kaizen

Kaizen also known as ***continuous improvement***, where small changes are made to the production process resulting in small improvements being made.

This system tends to be carried out on a regular basis as the changes made are often low cost and the improvements made tend to be small.

Lean Manufacturing using Just-in-Time (JIT) Systems

Flexible Manufacturing Cells

- Customers demand variety and customisation of products as well as specific quantities delivered at specific times, a lean manufacturer must remain flexible enough to serve its customers' needs.
- ***Manufacturing Cells*** allow manufacturers to provide their customers with the right product at the right time.
- They achieve this by grouping similar products into families that can be processed on the same equipment in the same sequence.

Lean Manufacturing using Just-in-Time (JIT) Systems

Flexible Manufacturing Cells

- A manufacturing cell is a group of workstations, machines or equipment arranged such that a product can be processed progressively from one workstation to another without having to wait for a batch to be completed and without additional handling between operations.
- Cell production methods provide flexibility as upgrades to processes can be performed relatively quickly and easily by shutting down one cell whilst another simultaneously opens and takes its place.
- This means that the entire production line does not have to be disrupted, which *reduces costs from stoppages or* ***'downtime'.***

Lean Manufacturing using Just-in-Time (JIT) Systems

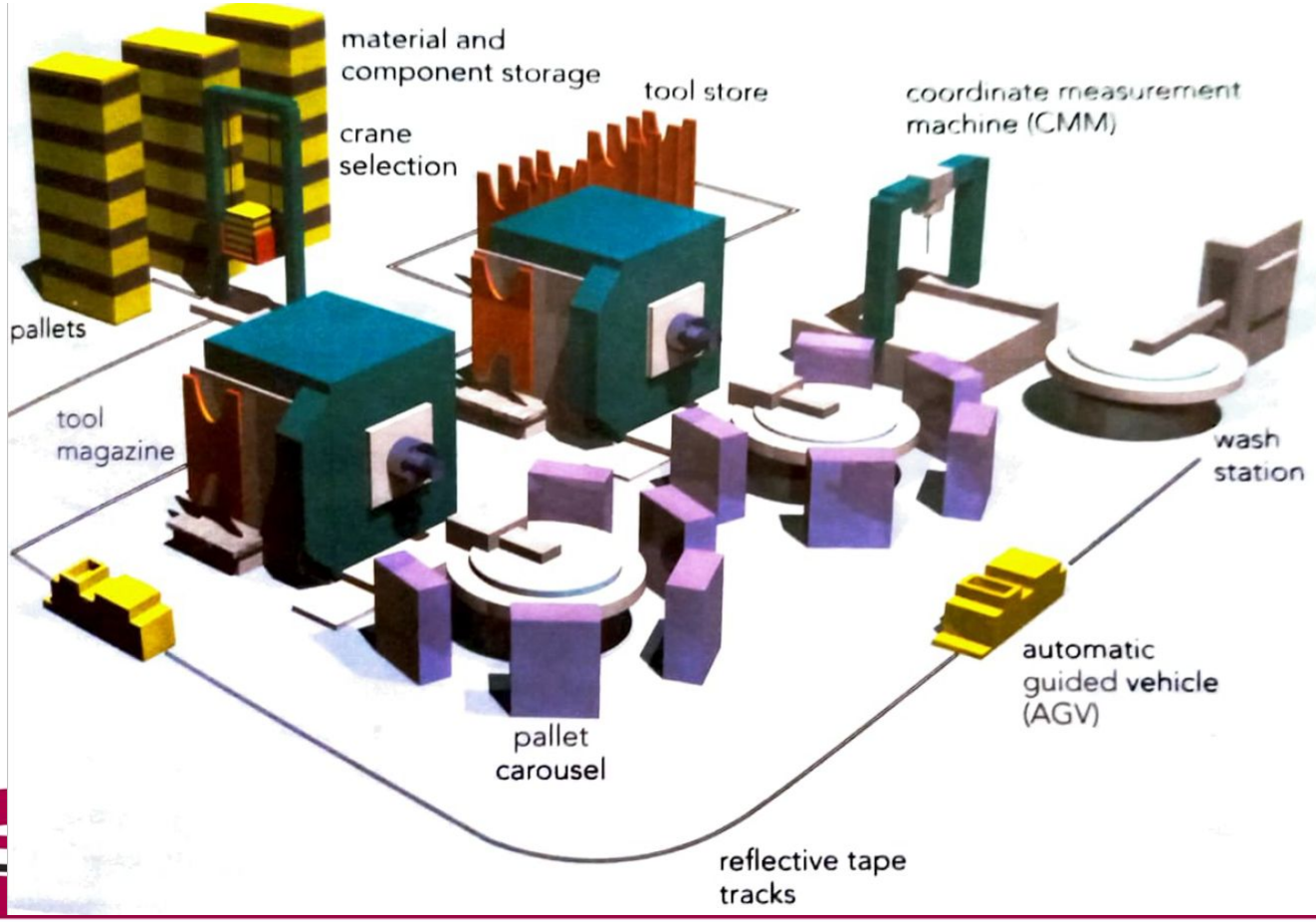
Flexible Manufacturing Cells

- Flexibility also occurs when a large variety of products are assembled from a range of similar components.
- For example, cellular manufacturing enables a large variety of cars to be produced using various combinations of customer options such as engine sizes, interior finishes etc.
- Typically, a manufacturing cell involves 3 – 12 workers and 5 – 15 workstations in a compact arrangement.
- However, manufacturing cells can be fully automated using CAM containing several CNC machines, computer aided quality control and automated materials-handling systems.

Lean Manufacturing using Just-in-Time (JIT) Systems

Flexible Manufacturing Cell

This is an ideal cell layout as it manufactures a narrow range of very similar products and is self-contained with all necessary equipment and resources. Materials sit in an initial queue when they enter the cell. Once processing begins, they move directly from process to process, resulting in a very fast throughput.



Video: JIT Production – Lean Production

An example of JIT in action today

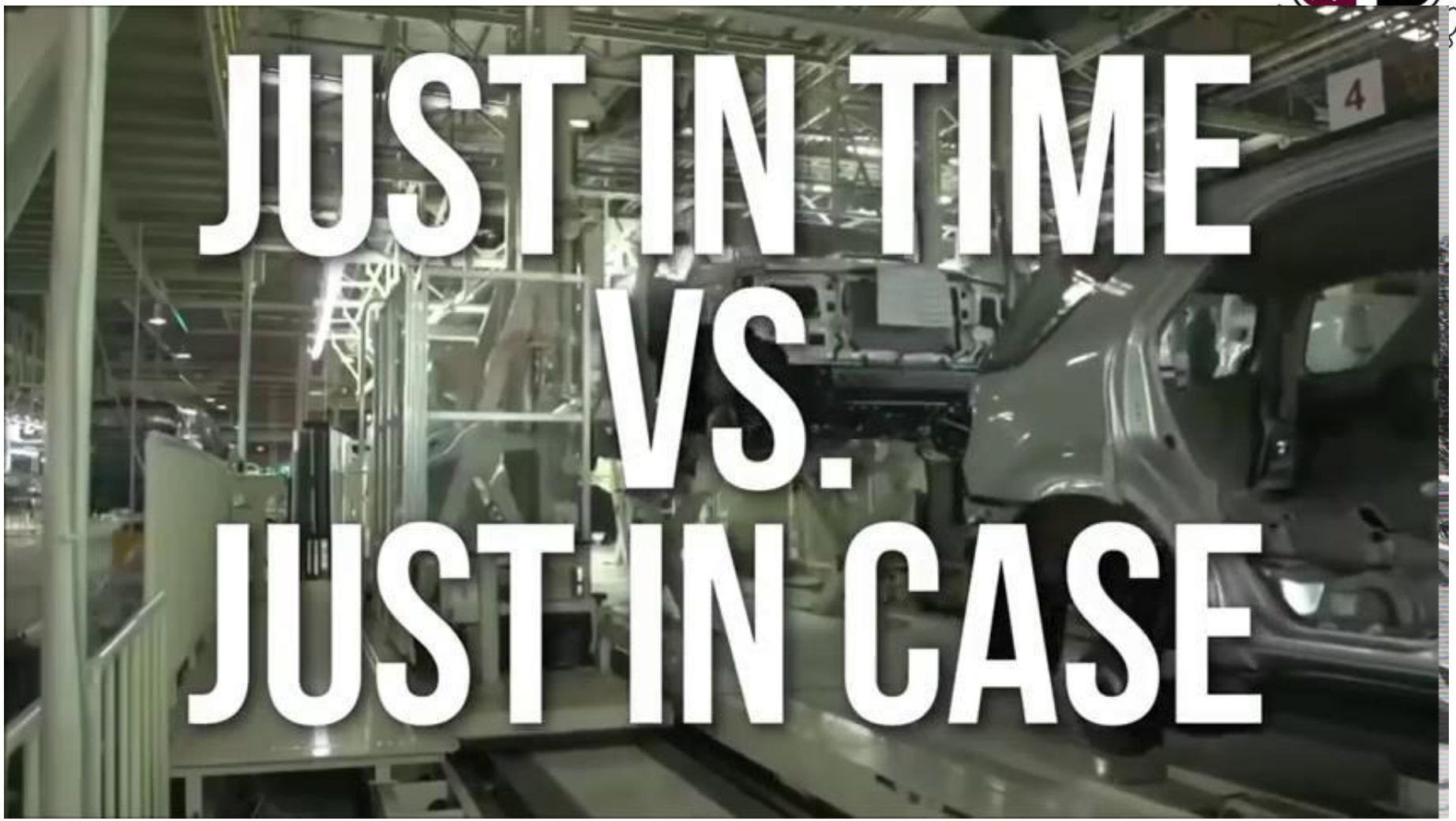
Just-in-time: The production system Brexit is set to sabotage



"Honda UK said that a delay of as little as 15 minutes to each truck that passes through Dover would add £850,000 to its annual running costs."

tutor2u

Video: JiT by Toyota



Now Test Yourself:

1. *Name* and *briefly explain* ***four*** key features of ***a just-in-time (JIT)*** manufacturing system and explain the key area for improvement that is targeted by a ***'lean' manufacturing*** approach to JIT.

2. *Quality control* and *efficiency* are key issues in modern manufacturing. ***Value*** and ***value stream*** are the first ***two*** *stages* of ***lean manufacturing***. *Explain* the ***three*** further stages of ***lean manufacturing***.